

Part A

- ① Find the velocity and acceleration of a particle whose position vector is $\vec{r}(t) = e^t \hat{i} + e^{-t} \hat{j}$ at $t=0$

Sol: $\frac{d\vec{r}}{dt} = e^t \hat{i} + e^{-t} \times -1 \hat{j} = \text{velocity} = \vec{v}$

$\therefore \vec{v}(0) = e^0 \hat{i} + e^{-0} \times -1 \hat{j} = \underline{\underline{\hat{i} - \hat{j}}}$

$\frac{d^2\vec{r}}{dt^2} = e^t \hat{i} + e^{-t} \times 1 \hat{j} = \text{acceleration} = \vec{a}$

$\therefore \vec{a}(0) = e^0 \hat{i} + e^{-0} \hat{j} = \underline{\underline{\hat{i} + \hat{j}}}$

- ② If C is the unit circle $x^2 + y^2 = 1$ oriented counter clockwise, find $\int_C x dx + y dy$

Sol: Parametric eqn. of $x^2 + y^2 = 1$ are $x = \cos t$ and $y = \sin t$ where $t: 0 \rightarrow 2\pi$

$\int_C x dx + y dy = \int_0^{2\pi} \cos t \cdot (-\sin t) dt + \sin t \cdot \cos t dt = 0$ $\begin{cases} dx = -\sin t dt \\ dy = \cos t dt \end{cases}$

- ③ Determine the sources & sinks of the vector field $\vec{F} = 2(x^3 - 2x) \hat{i} + 2(y^3 - 2y) \hat{j} + 2(z^3 - 2z) \hat{k}$

Sol: $\text{Div } \vec{F} = 2(3x^2 - 2) + 2(3y^2 - 2) + 2(3z^2 - 2)$
 $= 6x^2 + 6y^2 + 6z^2 - 12$

$$\text{Sources} \Rightarrow \operatorname{div} \vec{F} > 0 \Rightarrow 6x^2 + 6y^2 + 6z^2 - 12 > 0 \\ \Rightarrow x^2 + y^2 + z^2 > 2$$

$$\text{Sinks} \Rightarrow \operatorname{div} \vec{F} < 0 \Rightarrow 6x^2 + 6y^2 + 6z^2 - 12 < 0 \\ \Rightarrow x^2 + y^2 + z^2 < 2$$

- ④ Evaluate $\iint_{\sigma} xz \, ds$ where σ is the part of the plane $x+y+z=1$ that lies in the 1st octant.

Sol: Projecting the surface to xy -plane,

$$\iint_{\sigma} f(x, y, z) \, ds = \iint_R f(x, y, g(x, y)) \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1} \, dx \, dy$$

$$\therefore \iint_{\sigma} xz \, ds = \iint_R x(1-x-y) \sqrt{1+1+1} \, dx \, dy$$

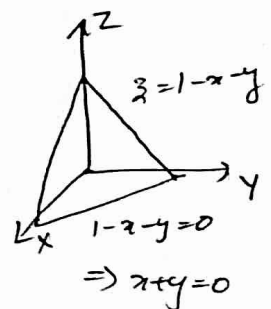
$$= \int_{y=0}^1 \int_{x=0}^{1-y} (x - x^2 - xy) \sqrt{3} \, dx \, dy$$

$$= \int_{y=0}^1 \left(\frac{x^2}{2} - \frac{x^3}{3} - \frac{x^2}{2} \right)_{x=0}^{1-y} \sqrt{3} \, dy$$

$$= \sqrt{3} \int_{y=0}^1 \left[\frac{(1-y)^2}{2} - \frac{(1-y)^3}{3} - \frac{y-2y^2+y^3}{2} \right] dy$$

$$= \sqrt{3} \left[\frac{1}{2} \frac{(1-y)^3}{3} - \frac{1}{3} \cdot \frac{(1-y)^4}{-4} - \frac{1}{2} \left(\frac{y^2}{2} - \frac{2y^3}{3} + \frac{y^4}{4} \right) \right]_{y=0}^1$$

$$= \frac{\sqrt{3}}{24}$$



$$z = 1 - x - y \\ \frac{\partial z}{\partial x} = -1, \quad \frac{\partial z}{\partial y} = -1$$

5. Solve the i.v.p $y'' + y' - 2y = 0$, $y(0) = 4$, $y'(0) = -5$

sol: $y'' + y' - 2y = 0 \Rightarrow \frac{d^2 y}{dx^2} + \frac{dy}{dx} - 2y = 0$

$$\Rightarrow (D^2 + D - 2)y = 0$$

A.E: $m^2 + m - 2 = 0 \Rightarrow \boxed{\text{OR}} \quad D^2 + D - 2 = 0 \Rightarrow D = 1, -2$

$$\therefore y = C_1 e^x + C_2 e^{-2x} \text{ --- (1)}$$

Given: $y(0) = 4$

Putting $x=0$ in (1), $y(0) = C_1 e^0 + C_2 e^0 = C_1 + C_2 \text{ --- (2)}$

$$\Rightarrow C_1 + C_2 = 4 \text{ --- (2)}$$

From (1); $y' = C_1 e^x - 2C_2 e^{-2x} \text{ ---}$

$$\therefore y'(0) = C_1 e^0 - 2C_2 e^0 = C_1 - 2C_2$$

Given: $y'(0) = -5 \quad \therefore C_1 - 2C_2 = -5 \text{ --- (3)}$

From (2) & (3): $C_1 + C_2 = 4$

$$C_1 - 2C_2 = -5$$

Subtracting: $3C_2 = 9 \Rightarrow \underline{\underline{C_2 = 3}}$

$$\therefore \underline{\underline{C_1 = 1}}$$

Solution (1) becomes,

$$y = \underline{\underline{e^x + 3e^{-2x}}}$$

⑥ Solve: $y'''' + 10y'' + 9y = 0$

Sol: $\frac{d^4 y}{dx^4} + 10 \frac{d^2 y}{dx^2} + 9y = 0 \Rightarrow (D^4 + 10D^2 + 9)y = 0$

A.E: $D^4 + 10D^2 + 9 = 0 \Rightarrow D = \pm i, \pm 3i$

$\therefore$ Solution is, $y = C_1 \cos x + C_2 \sin x + C_3 \cos 3x + C_4 \sin 3x$

⑦ Find the Laplace transform of $e^{-3t} \cos 2t$.

Sol: $L[\cos 2t] = \frac{s}{s^2 + 2^2} = \frac{s}{s^2 + 4}$

By 1st shifting theorem,

$L[e^{-3t} \cos 2t] = \frac{s+3}{(s+3)^2 + 4}$

⑧ Find the inverse Laplace transform of $\frac{1}{(s+a)(s+b)}$

Sol: $\frac{1}{(s+a)(s+b)} = \frac{A}{s+a} + \frac{B}{s+b} \Rightarrow 1 = A(s+b) + B(s+a)$

$s = -b \Rightarrow 1 = A \cdot 0 + B(a-b) \Rightarrow 1 = B(a-b) \Rightarrow B = \frac{1}{a-b}$

$s = -a \Rightarrow 1 = A(b-a) + 0 \Rightarrow 1 = A(b-a) \Rightarrow A = \frac{1}{b-a}$

$L^{-1}\left[\frac{1}{(s+a)(s+b)}\right] = \frac{1}{b-a} L^{-1}\left(\frac{1}{s+a}\right) + \frac{1}{a-b} L^{-1}\left(\frac{1}{s+b}\right)$
 $= \frac{1}{b-a} e^{-at} + \frac{1}{a-b} e^{-bt}$

- ⑨ Find the Fourier cosine transform of $f(x) = \begin{cases} x, & 0 < x < a \\ 0, & x > a \end{cases}$

Sol:

$$\begin{aligned} \hat{f}_c(\omega) &= \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \cos \omega x \, dx \\ &= \sqrt{\frac{2}{\pi}} \int_0^a x \cos \omega x \, dx \\ &= \sqrt{\frac{2}{\pi}} \left[x \cdot \frac{\sin \omega x}{\omega} - 1 \cdot \frac{\cos \omega x}{\omega^2} \right]_{x=0}^a \\ &= \sqrt{\frac{2}{\pi}} \left\{ \left[a \frac{\sin \omega a}{\omega} + \frac{\cos \omega a}{\omega^2} \right] - \left[0 + \frac{1}{\omega^2} \right] \right\} \\ &= \sqrt{\frac{2}{\pi}} \left[a \frac{\sin \omega a}{\omega} + \frac{\cos \omega a - 1}{\omega^2} \right] \end{aligned}$$

- ⑩ Using Fourier sine integral, show that

$$\int_0^{\infty} \frac{1 - \cos \pi \omega}{\omega} \sin x \omega \, d\omega = \begin{cases} \frac{\pi}{2} & \text{if } 0 < x < \pi \\ 0 & \text{if } x > \pi \end{cases}$$

Sol: Let $f(x) = \begin{cases} \frac{\pi}{2}, & 0 < x < \pi \\ 0, & x > \pi \end{cases}$

Fourier sine integral of $f(x)$ is

$$f(x) = \int_0^{\infty} B(\omega) \sin \omega x \, d\omega \quad \text{where} \quad \textcircled{1}$$

$$\begin{aligned} B(\omega) &= \frac{2}{\pi} \int_0^{\infty} f(x) \sin \omega x \, dx = \frac{2}{\pi} \int_0^{\pi} \frac{\pi}{2} \sin \omega x \, dx = \left[\frac{\cos \omega x}{\omega} \right]_0^{\pi} \\ &= -\frac{\cos \omega \pi}{\omega} - \frac{\cos 0}{\omega} = \frac{1 - \cos \omega \pi}{\omega} \quad \text{Substituting } B(\omega) \end{aligned}$$

in $\textcircled{1}$ we get the answer.

Module 1

⑪

a. Find the directional derivative of $\phi = x^3 - yx^2 + z^2$ at $(1, 1, 1)$ in the direction of $\vec{a} = 2\hat{i} - \hat{j} + 2\hat{k}$. Also find the maximum directional derivative.

sol.

$$D_{\vec{a}} \bar{f} = \nabla \bar{f} \cdot \frac{\vec{a}}{\|\vec{a}\|}$$

$$= \nabla [x^3 - yx^2 + z^2] \cdot \frac{2\hat{i} - \hat{j} + 2\hat{k}}{\sqrt{2^2 + (-1)^2 + 2^2}}$$

$$= \hat{i} [3x^2 - 2yx] + \hat{j} [-x^2] + \hat{k} [x^3 + 2z] \cdot \left[\frac{2\hat{i} - \hat{j} + 2\hat{k}}{3} \right]$$

$$\left[D_{\vec{a}} \bar{f} \right]_{(1,1,1)} = \hat{i} [3-2] + \hat{j} [-1] + \hat{k} [1+2] \cdot \left[\frac{2\hat{i} - \hat{j} + 2\hat{k}}{3} \right]$$

$$= (\hat{i} - \hat{j} + 3\hat{k}) \cdot \left(\frac{2\hat{i} - \hat{j} + 2\hat{k}}{3} \right)$$

$$= \frac{2+1+6}{3} = \frac{9}{3} = \underline{\underline{3}}$$

$$\text{Maximum D.D} = |\nabla \phi|_{(1,1,1)}$$

$$= |\hat{i} - \hat{j} + 3\hat{k}|$$

$$= \sqrt{1^2 + (-1)^2 + 3^2}$$

$$= \underline{\underline{\sqrt{11}}}$$

11 (b) Show that the vector field $\vec{F} = xy^2 \hat{i} + x^2y \hat{j}$ is conservative and find ϕ such that $\vec{F} = \nabla\phi$. Hence evaluate

$$\int_{(1,2)}^{(2,4)} xy^2 dx + x^2y dy$$

Sol: $\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy^2 & x^2y & 0 \end{vmatrix} = \hat{i}[0-0] - \hat{j}[0-0] + \hat{k}[2xy-2xy] = 0$
 $\Rightarrow \vec{F}$ is conservative.
 $\Rightarrow \vec{F} = \nabla\phi$

$$\therefore xy^2 \hat{i} + x^2y \hat{j} = \nabla\phi = \hat{i} \frac{\partial\phi}{\partial x} + \hat{j} \frac{\partial\phi}{\partial y}$$

$$\Rightarrow \begin{cases} \frac{\partial\phi}{\partial x} = xy^2 \rightarrow \phi = \frac{x^2}{2}y^2 + g(y) \text{ for not having } x \\ \frac{\partial\phi}{\partial y} = x^2y \rightarrow \phi = x^2 \frac{y^2}{2} + f(x) \text{ for not having } y \end{cases}$$

$$\therefore \underline{\underline{\phi = \frac{x^2y^2}{2} + C}}$$

$$\begin{aligned} \int_{(1,2)}^{(2,4)} xy^2 dx + x^2y dy &= \phi(2,4) - \phi(1,2) \\ &= \left[\frac{2^2 \cdot 4^2}{2} + C \right] - \left[\frac{1^2 \cdot 2^2}{2} + C \right] \\ &= 32 - 2 \\ &= \underline{\underline{30}} \end{aligned}$$

(12)

a Find the parametric equation of the tangent line to the graph $\vec{r}(t) = t^2 \hat{i} - \frac{1}{t+3} \hat{j} + (4-t^2) \hat{k}$ at $(4, -1, 0)$

sol:

$$\vec{r}(t) = \vec{r}(t_0) + t \vec{r}'(t_0) \quad ; \quad t_0(4, -1, 0) \Rightarrow \text{at } t = -2$$

$$\vec{r}(t_0) = \vec{r}(-2)$$

$$= 4\hat{i} - \frac{1}{-2+3} \hat{j} + 4 - (-2)^2 \hat{k}$$

$$= 4\hat{i} - \hat{j}$$

$$\vec{r}'(t) = 2t \hat{i} - \frac{-1}{(t+3)^2} \hat{j} - 2t \hat{k}$$

$$\vec{r}'(-2) = -4\hat{i} + \hat{j} + 4\hat{k}$$

$$\therefore \vec{r}(t) = 4\hat{i} - \hat{j} + t[-4\hat{i} + \hat{j} + 4\hat{k}]$$

$$= (4-4t)\hat{i} + (-1+t)\hat{j} + 4t\hat{k}$$

parametric equ. are

$$x(t) = 4-4t$$

$$y(t) = -1+t$$

$$\underline{\underline{z(t) = 4t}}$$

$$x(t) = t^2$$

$$y(t) = \frac{-1}{t+3}$$

$$z(t) = 4-t^2$$

$$y(t) = \frac{-1}{t+3} = -1$$

$$\Rightarrow -1 = t+3$$

$$\Rightarrow -1-3 = t = -2$$

$$4-t^2=0 \Rightarrow t^2=4$$

$$\Rightarrow t = \pm 2$$

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- ⑥ Using line integral evaluate $\int_C x^2 y dx + xy dy$ where C is the triangular path connecting $(0,0)$, $(1,0)$ and $(1,2)$ in the +ve direction.

Sol:

$$\int_C x^2 y dx + xy dy = \int_{C_1} x^2 y dx + xy dy + \int_{C_2} x^2 y dx + xy dy + \int_{C_3} x^2 y dx + xy dy$$

$$= \int_{C_1} 0 + \int_{y=0}^2 y dy + \int_{y=2}^0 \left(\frac{y}{2}\right)^2 y \cdot \frac{dy}{2} + \frac{y}{2} \cdot y dy$$

$$= \left(\frac{y^2}{2}\right)_0^2 + \int_{y=2}^0 \left(\frac{y^3}{8} + \frac{y^2}{2}\right) dy$$

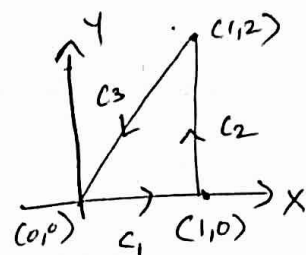
$$= 2 + \left[\frac{y^4}{32} + \frac{y^3}{6} \right]_2^0$$

$$= 2 + \left[0 - \left(\frac{16}{32} + \frac{8}{6} \right) \right]$$

$$= 2 - \frac{1}{2} - \frac{4}{3}$$

$$= \frac{12 - 3 - 8}{6}$$

$$= \underline{\underline{\frac{1}{6}}}$$



$$C_1: y=0; x: 0 \rightarrow 1$$

$$C_2: x=1, y: 0 \rightarrow 2$$

$$C_3: x=\frac{y}{2}, y: 2 \rightarrow 0$$

Equ- of C_3

$$\begin{matrix} (0,0) & (1,2) \\ (x_1, y_1) & (x_2, y_2) \end{matrix}$$

$$\frac{x-0}{1-0} = \frac{y-0}{2-0}$$

$$\underline{\underline{x = y/2}}$$

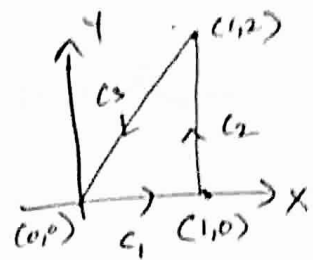
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- (b) Using line integral evaluate $\int_C x^2 y dx + xy dy$ where C is the triangular path connecting $(0,0)$, $(1,0)$ and $(1,2)$ in the +ve direction.

Sol.

$$\int_C x^2 y dx + xy dy = \int_{C_1} x^2 y dx + xy dy$$

$$+ \int_{C_2} x^2 y dx + xy dy + \int_{C_3} x^2 y dx + xy dy$$



$$C_1: y=0, x: 0 \rightarrow 1$$

$$C_2: x=1, y: 0 \rightarrow 2$$

$$C_3: x=\frac{y}{2}, y: 2 \rightarrow 0$$

$$= \int_{C_1} 0 + \int_{y=0}^2 y dy + \int_{y=2}^0 \left(\frac{y}{2}\right)^2 y \cdot \frac{dy}{2} + \frac{y}{2} \cdot y dy$$

$$= \left(\frac{y^2}{2}\right)_0^2 + \int_{y=2}^0 \left(\frac{y^3}{8} + \frac{y^2}{2}\right) dy$$

$$= 2 + \left[\frac{y^4}{32} + \frac{y^3}{6} \right]_2^0$$

$$= 2 + \left[0 - \left(\frac{16}{32} + \frac{8}{6} \right) \right]$$

$$= 2 - \frac{1}{2} - \frac{4}{3}$$

$$= \frac{12 - 3 - 8}{6}$$

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Equ. of C_3

$$\begin{matrix} (0,0) & (1,2) \\ (x_1, y_1) & (x_2, y_2) \end{matrix}$$

$$\frac{x-0}{1-0} = \frac{y-0}{2-0}$$

$$\underline{\underline{x = y/2}}$$

Module II

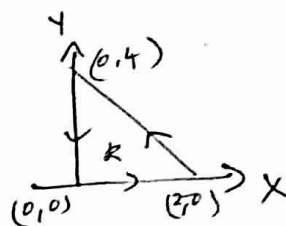
⑬

- a. Use Green's theorem to evaluate $\int_C \log(1+y) dx - \frac{xy}{1+y} dy$ where C is the triangle with vertices $(0,0), (2,0), (0,4)$

sol: $\int_C f dx + g dy = \iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dA \rightarrow \text{Green's theorem}$

$$f = \log(1+y), \quad g = \frac{-xy}{1+y}$$

$$\frac{\partial f}{\partial y} = \frac{1}{1+y}, \quad \frac{\partial g}{\partial x} = \frac{-y}{1+y}$$



Using Green's theorem,

$$\int_C \log(1+y) dx - \frac{xy}{1+y} dy = \iint_R \left(\frac{-y}{1+y} - \frac{1}{1+y} \right) dA$$

$$= - \iint_R dA = - \text{Area of } \Delta^k$$

$$= -\frac{1}{2} \times 2 \times 4 = \underline{\underline{-4}}$$

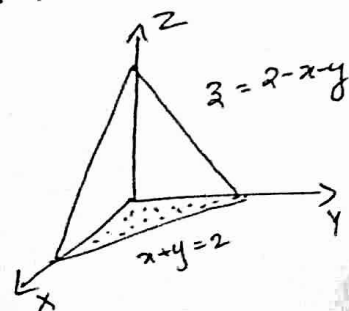
⑭

- b. Use divergence theorem to find the outward flux of the vector field $\vec{F} = (2x+y^2)\hat{i} + xy\hat{j} + (xy-2z)\hat{k}$ across the surface of the tetrahedron bounded by $x+y+z=2$ and the coordinate planes.

sol: Outward flux $\phi = \iint_S \vec{F} \cdot \hat{n} ds$

$$= \iiint_V \nabla \cdot \vec{F} dV$$

$$= \iiint_V 2 + x + -2 dV$$



$$\phi = \iiint_G x \, dz \, dy \, dx$$

$$= \int_{x=0}^2 \int_{y=0}^{2-x} \int_{z=0}^{2-x-y} x \, dz \, dy \, dx$$

$$= \int_{x=0}^2 \int_{y=0}^{2-x} x(2-x-y) \, dy \, dx$$

$$= \int_{x=0}^2 \int_{y=0}^{2-x} (2x - x^2 - xy) \, dy \, dx$$

$$= \int_{x=0}^2 \left(2xy - x^2y - \frac{xy^2}{2} \right)_{y=0}^{2-x} dx$$

$$= \int_{x=0}^2 \left[2x(2-x) - x^2(2-x) - \frac{x}{2}(2-x)^2 \right] dx$$

$$= \int_{x=0}^2 \left(4x - 4x^2 + x^3 - 2x + \frac{x^3}{2} + 2x^2 \right) dx$$

$$= \left(4\frac{x^2}{2} - 4\frac{x^3}{3} + \frac{x^4}{4} - 2\frac{x^2}{2} - \frac{x^4}{8} + 2\frac{x^3}{3} \right)_0^2$$

$$= 2 \times \frac{4}{2} - \frac{4}{3} \times 8 + \frac{2^4}{4} - 4 - \frac{2^4}{8} + \frac{2}{3} \times 8$$

$$= \frac{2}{3}$$

14) Find the flux of the vector field $\vec{F}(x, y, z) = x\hat{k}$ across the surface, the portion of the paraboloid $z = x^2 + y^2$ below the plane $z = 2y$ oriented by downward unit normal.

Sol: $\text{flux}, \phi = \iint_S \vec{F} \cdot \hat{n} \, ds$

$$= \iiint_V \nabla \cdot \vec{F} \, dV \quad \text{Using divergence theorem}$$

$$\boxed{\nabla \cdot \vec{F} = 0}$$

$$= 0$$

OR

$$\phi = \iint_S \vec{F} \cdot \hat{n} \, ds$$

$$= \iint_S (x\hat{k}) \cdot (2x\hat{i} + 2y\hat{j} - \hat{k}) \, dA$$

$$= \iint_S -x \, dA$$

$$= \int_0^\pi \int_0^{2\sin\theta} r \cos\theta \cdot r \, dr \, d\theta$$

$\theta=0 \quad r=0$

$$= 0$$

$$x^2 + y^2 = r^2$$

$$z = x^2 + y^2$$

$$\hat{n} = \frac{\partial z}{\partial x} \hat{i} + \frac{\partial z}{\partial y} \hat{j} - \hat{k}$$

$$= 2x\hat{i} + 2y\hat{j} - \hat{k}$$

(14)

b Use Stokes theorem to evaluate $\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = (z-y)\hat{i} + (z+x)\hat{j} - (x+y)\hat{k}$ and C is the boundary of the paraboloid $z = 9 - x^2 - y^2$ above the xy -plane with upward orientation.

Sol: Stokes theorem: $\int_C \vec{F} \cdot d\vec{r} = \iint_S \text{Curl } \vec{F} \cdot \hat{n} \, ds$

$$\begin{aligned} \text{Curl } \vec{F} = \nabla \times \vec{F} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ z-y & z+x & -x-y \end{vmatrix} \\ &= \hat{i}(-1-1) - \hat{j}(-1-1) + \hat{k}(1-1) \\ &= -2\hat{i} + 2\hat{j} + 2\hat{k} \end{aligned}$$

$$\begin{aligned} \hat{n} &= -\frac{\partial z}{\partial x} \hat{i} - \frac{\partial z}{\partial y} \hat{j} + \hat{k} \\ &= -(-2x) \hat{i} - (-2y) \hat{j} + \hat{k} \\ &= 2x\hat{i} + 2y\hat{j} + \hat{k} \end{aligned}$$

$$\begin{aligned} \therefore \int_C \vec{F} \cdot d\vec{r} &= \iint_R (-2\hat{i} + 2\hat{j} + 2\hat{k}) \cdot (2x\hat{i} + 2y\hat{j} + \hat{k}) \, dA \\ &= \iint_R (-4x + 4y + 2) \, dA \\ &= \int_{\theta=0}^{2\pi} \int_{r=0}^3 (-4r\cos\theta + 4r\sin\theta + 2) r \, dr \, d\theta \\ &= 18\pi \end{aligned}$$

MODULE III

(15)

(a) Solve using the method of undetermined coefficients $y'' - 4y' - 5y = 4 \cos 2x$.

(b) Soln:-

$$y'' - 4y' - 5y = 4 \cos 2x \longrightarrow \textcircled{1}$$

Annihilary equation is

$$m^2 - 4m - 5 = 0 \Rightarrow m = 5, -1$$

Hence Complementary Function is

$$y_h = C_1 e^{5x} + C_2 e^{-x}$$

Now $r(x) = 4 \cos 2x$.

Choose PI as $y_p = A \cos 2x + B \sin 2x$

$$y_p' = -2A \sin 2x + 2B \cos 2x$$

$$y_p'' = -4A \cos 2x - 4B \sin 2x$$

Substituting y_p, y_p', y_p'' in $\textcircled{1}$,

① $\Rightarrow$

$$\begin{aligned} & (-4A \cos 2x - 4B \sin 2x) - 4(-2A \sin 2x \\ & + 2B \cos 2x) - 5(A \cos 2x + B \sin 2x) \\ & = 4 \cos 2x \end{aligned}$$

$$\begin{aligned} \cos 2x [-4A - 8B] + \sin 2x [-4B + 8A - 5A] \\ = 4 \cos 2x \end{aligned}$$

$$\Rightarrow -9A - 8B = 4$$

$$-9B + 8A = 0$$

$$\text{Solving } A = \frac{-36}{145}, B = \frac{-32}{145}$$

$$\Rightarrow y_p = \frac{-36}{145} \cos 2x - \frac{32}{145} \sin 2x$$

$\therefore$ Complete solution is, $y = y_h + y_p$

$$y = C_1 e^{5x} + C_2 e^{-x} + \left(\frac{-36}{145} \cos 2x - \frac{32}{145} \sin 2x \right)$$

$$= C_1 e^{5x} + C_2 e^{-x} - \left(\frac{36 \cos 2x + 32 \sin 2x}{145} \right)$$

(b) Solve using the method of variation of parameters $y'' + y = \operatorname{cosec} x$.

Soln:-

$$y'' + y = \operatorname{cosec} x.$$

To find y_h

Auxiliary equation is,

$$m^2 + 1 = 0 \Rightarrow m = \pm i$$

$$\therefore \boxed{y_h = C_1 \cos x + C_2 \sin x}$$

To find y_p

$$y_p(x) = -y_1 \int \frac{y_2}{W} r(x) dx + y_2 \int \frac{y_1}{W} r(x) dx.$$

$$y_1 = \cos x, \quad y_2 = \sin x, \quad y_1' = -\sin x, \quad y_2' = \cos x$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = \cos^2 x + \sin^2 x = \underline{\underline{1}}$$

We have $r(x) = \operatorname{cosec} x$

$$\begin{aligned}
 y_p(x) &= -\cos x \int \frac{\sin x}{1} \cos x dx + \sin x \int \frac{\cos x}{1} \cos x dx \\
 &= -\cos x \int dx + \sin x \int \cot x dx \\
 &= -\cos x (x) + \sin x (\log |\sin x|) + C
 \end{aligned}$$

Complete solution is

$$y = y_h + y_p$$

$$\Rightarrow y = (C_1 \cos x + C_2 \sin x) + \sin(\log |\sin x|) - (\cos x)x$$

16) Solve using the method of undetermined coefficients

$$y'' - 7y' + 12y = e^{2x}, \quad y(0) = 1, \quad y'(0) = 2$$

Soln:-

$$y'' - 7y' + 12y = e^{2x} \rightarrow (1)$$

To find $y_h(x)$

Auxiliary equation is

$$m^2 - 7m + 12 = 0 \Rightarrow m = 3, 4$$

$$\therefore \boxed{y_h(x) = C_1 e^{3x} + C_2 e^{4x}}$$

To find $y_p(x)$

$$\text{Choose } y_p(x) = A e^{2x}.$$

$$\Rightarrow y_p' = 2A e^{2x}$$

$$y_p'' = 4A e^{2x}.$$

Substituting in ①,

$$4A e^{2x} - 7(2A e^{2x}) + 12A e^{2x} = e^{2x}$$

$$\Rightarrow 2A = 1 \Rightarrow \boxed{A = \frac{1}{2}}$$

$$\therefore \boxed{y_p(x) = \frac{1}{2} e^{2x}}.$$

General solution is $y = y_h + y_p$

$$\Rightarrow \underline{\underline{y = (C_1 e^{3x} + C_2 e^{4x}) + \frac{1}{2} e^{2x} \rightarrow \textcircled{2}}}}$$

To solve IVP

$$y(0) = 1, \quad y'(0) = 2.$$

$$\underline{y(0) = 1} \Rightarrow$$

$$\textcircled{2} \Rightarrow 1 = c_1 + c_2 + \frac{1}{2}$$

$$\Rightarrow c_1 + c_2 = \frac{1}{2} \rightarrow \textcircled{3}$$

$$y'(x) = 3c_1 e^{3x} + 4c_2 e^{4x} + e^{2x} \rightarrow \textcircled{4}$$

$$\text{Now } y'(0) = 2 \text{ in } \textcircled{4}$$

$$\Rightarrow 2 = 3c_1 + 4c_2 + 1$$

$$\Rightarrow 3c_1 + 4c_2 = 1 \rightarrow \textcircled{5}$$

Solving $\textcircled{3}$ and $\textcircled{5}$

$$\boxed{\begin{matrix} c_1 = 1 \\ c_2 = -\frac{1}{2} \end{matrix}}$$

$$\begin{array}{r} c_1 + c_2 = \frac{1}{2} \\ 3c_1 + 4c_2 = 1 \\ \hline \Rightarrow c_2 = -\frac{1}{2} \\ c_1 = 1 \end{array}$$

Hence solution of IVP is

$$y = (c_1 e^{3x} + c_2 e^{4x}) + \frac{1}{2} e^{2x}$$

$$\Rightarrow y = e^{3x} - \frac{1}{2} e^{4x} + \frac{1}{2} e^{2x}$$

(b) solve the boundary value problem

$$x^2 y'' - 3x y' + 4y = 0, \quad y(1) = 0, \quad y'(1) = 1$$

Soln:- $x^2 y'' - 3x y' + 4y = 0 \rightarrow \textcircled{1}$

We have Euler - Cauchy equation

$$x^2 y'' + a x y' + b y = 0.$$

Comparing

$$\Rightarrow a = -3, \quad b = 4$$

Auxiliary equation is,

$$m^2 + (a-1)m + b = 0$$

$$\Rightarrow m^2 - 4m + 4 = 0 \Rightarrow m = 2, 2$$

Solution is

$$\boxed{y = (C_1 + C_2 \log x) x^2} \rightarrow \textcircled{2}$$

Applying $y(1) = 0$ in $\textcircled{2}$,

$$0 = (C_1 + C_2 \log 1) \Rightarrow \boxed{C_1 = 0}$$

$$\text{Now } y' = (C_1 + C_2 \log x) 2x + x^2 \left(C_2 \left(\frac{1}{x} \right) \right) \rightarrow \textcircled{3}$$

$$\underline{y'(1) = 1 \text{ in } (3)}$$

$$1 = (c_1 + c_2 \log 1) 2 + c_2$$

$$1 = 2c_1 + c_2 \Rightarrow \boxed{c_2 = 1}$$

General solution is

$$\underline{\underline{y = (\log x) x^2}}$$

(17) (a)

MODULE IV

Using convolution theorem, find the Laplace inverse of $\frac{2}{(s^2+1)(s^2+25)}$.

Soln:-

$$\frac{2}{(s^2+1)(s^2+25)} = \frac{2}{s^2+1} \cdot \frac{1}{s^2+25}$$

$$F(s) = \frac{2}{s^2+1} \Rightarrow L^{-1}\left(\frac{2}{s^2+1}\right) = 2 \sin t = f(t)$$

$$G(s) = \frac{1}{s^2+25} \Rightarrow L^{-1}\left(\frac{1}{s^2+25}\right) = \frac{1}{5} \sin 5t = g(t)$$

By convolution theorem,

$$L^{-1}[F(s) \cdot G(s)] = f * g = \int_0^t f(u) g(t-u) du.$$

$$\begin{aligned}
\Rightarrow L^{-1} \left[\frac{2}{(s^2+1)(s^2+25)} \right] &= \int_0^t 2 \sin u \cdot \frac{1}{5} \sin 5(t-u) du \\
&= \frac{1}{5} \int_0^t 2 \sin u \cdot \sin (5t-5u) du \\
&= \frac{1}{5} \int_0^t \cos (6u-5t) - \cos (5t-4u) du \\
&= \frac{1}{5} \left[\frac{\sin (6u-5t)}{6} - \frac{\sin (5t-4u)}{-4} \right]_0^t \\
&= \frac{1}{5} \left[\left(\frac{\sin (6t-5t)}{6} + \frac{\sin (5t-4t)}{4} \right) - \left(\frac{\sin (5t)}{6} + \frac{\sin 5t}{4} \right) \right] \\
&= \frac{1}{5} \left[\frac{\sin t}{6} + \frac{\sin t}{4} + \frac{\sin 5t}{6} - \frac{\sin 5t}{4} \right] \\
&= \frac{1}{5} \left[\sin t \left(\frac{2+3}{12} \right) + \sin 5t \left(\frac{2-3}{12} \right) \right] \\
&= \frac{1}{5} \left[\frac{5 \sin t}{12} - \frac{\sin 5t}{12} \right] \\
&= \frac{1}{60} [5 \sin t - \sin 5t]
\end{aligned}$$

(b) Using Laplace Transform solve

$$y'' + 2y' + y = e^{-t}, \quad y(0) = 0, \quad y'(0) = 1$$

Soln:-

$$L(y'' + 2y' + y) = L(e^{-t})$$

$$\Rightarrow [s^2 L(y) - s y(0) - y'(0)] + 2[s L(y) - y(0)] + L(y) = \frac{1}{s+1}$$

$$\Rightarrow s^2 L(y) - 1 + 2s L(y) + L(y) = \frac{1}{s+1}$$

$$\Rightarrow L(y) = \frac{s+2}{(s+1)^3} = \frac{s+1+1}{(s+1)^3}$$

$$y(t) = L^{-1}\left[\frac{s+1}{(s+1)^3}\right] + L^{-1}\left[\frac{1}{(s+1)^3}\right]$$

$$= L^{-1}\left[\frac{1}{(s+1)^2}\right] + L^{-1}\left[\frac{1}{(s+1)^3}\right]$$

$$= \underline{\underline{e^{-t} t + \frac{e^{-t} t^2}{2!}}}$$

(18) (a) Express in terms of unit step function and hence find the Laplace transform of

$$f(t) = \begin{cases} t^2, & 0 < t < 2 \\ t-1, & 2 < t < 3 \\ 7, & t > 3 \end{cases}$$

Soln:-

$$t^2 = t^2 \times 1, \quad 0 < t < 2$$

$$= t^2 [1 - 0]$$

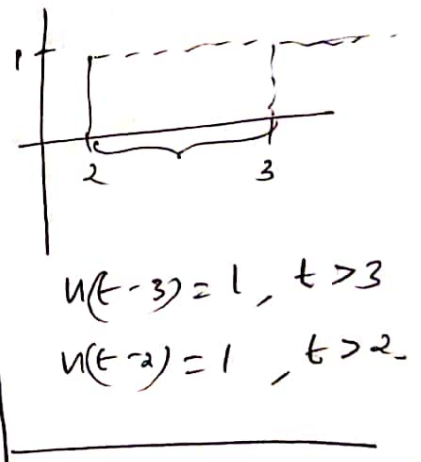
$$= t^2 [1 - u(t-2)]$$

$$t-1 = (t-1) [u(t-2) - u(t-3)]$$

$$7 = 7 \times 1$$

$$= 7 u(t-3)$$

$$\boxed{u(t-a) = \begin{cases} 0, & t < a \\ 1, & t > a \end{cases}}$$



$$\therefore f(t) = t^2 [1 - u(t-2)] + (t-1) [u(t-2) - u(t-3)] + 7 u(t-3)$$

$$\therefore L[f(t)] = L[t^2 - t^2 u(t-2)] + L[(t-1) u(t-2)] - L[(t-1) u(t-3)] + 7 L[u(t-3)]$$

$$= L(t^2) - L[t^2 u(t-2)] + L[(t-2+1)u(t-2)] \\ - L[(t-3+2)u(t-3)] + 7 L[u(t-3)]$$

$$= L(t^2) - L[t^2 u(t-2)] + L[(t-2)u(t-2)] + L[u(t-2)] \\ - L(t-3)u(t-3) - 2L[u(t-3)] + 7 L[u(t-3)]$$

$$\begin{aligned} &L[t^2 u(t-2)] \\ &= (-1)^2 \frac{d^2}{ds^2} \left[\frac{e^{-2s}}{s} \right] \\ &= e^{-2s} \left[\frac{4}{s} + \frac{4}{s^2} + \frac{2}{s^3} \right] \end{aligned}$$

$$= \frac{2}{s^3} - e^{-2s} \left[\frac{4}{s} + \frac{4}{s^2} + \frac{2}{s^3} \right] + e^{-2s} \left[\frac{1}{s^2} \right] \\ + e^{-2s} - e^{-3s} \left(\frac{1}{s^2} \right) - 2e^{-3s} + 7e^{-3s}$$

$$= \frac{2}{s^3} + \frac{3}{s} e^{-2s} - e^{-3s} \left[\frac{1}{s^2} + \frac{9}{s} \right]$$

(b) Find the inverse transform of $\frac{s^2+2}{s(s^2+9)}$

Soln:-

$$\frac{s^2+2}{s(s^2+9)} = \frac{A}{s} + \frac{Bs+C}{s^2+9}$$

$$\Rightarrow s^2 + 2 = A(s^2 + 9) + (Bs + C)s$$

$$\Rightarrow A = \frac{2}{9}, B = \frac{7}{9}, C = 0.$$

$$L^{-1} \left[\frac{s^2 + 2}{s(s^2 + 9)} \right] = L^{-1} \left[\frac{\frac{2}{9}}{s} \right] + L^{-1} \left[\frac{\frac{7}{9}s}{s^2 + 9} \right]$$

$$= \frac{2}{9} + \frac{7}{9} \cos 3t.$$

MODULE V

(19) (a)

Find the Fourier transform of

$$f(x) = \begin{cases} |x|, & -1 < x < 1 \\ 0 & \text{otherwise.} \end{cases}$$

Soln:-

$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx.$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-1}^1 |x| e^{-i\omega x} dx.$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-1}^1 |x| (\cos \omega x - i \sin \omega x) dx.$$

$$\begin{aligned}
&= \frac{1}{\sqrt{2\pi}} 2 \int_0^1 x \cos \omega x \, dx \quad \left| \begin{array}{l} x > 0, |x| = x \\ \cos \omega x \rightarrow \text{even} \\ \sin \omega x \rightarrow \text{odd} \end{array} \right. \\
&= \sqrt{\frac{2}{\pi}} \left[x \cdot \frac{\sin \omega x}{\omega} - \left(\frac{\cos \omega x}{\omega^2} \right) \right]_0^1 \\
&= \sqrt{\frac{2}{\pi}} \left[\frac{\sin \omega}{\omega} + \frac{\cos \omega}{\omega^2} - \left(0 + \frac{1}{\omega^2} \right) \right] \\
&= \sqrt{\frac{2}{\pi}} \left[\frac{\cos \omega + \omega \sin \omega - 1}{\omega^2} \right]
\end{aligned}$$

(b) Find the Fourier Integral of

$$f(x) = \begin{cases} \pi - x, & 0 < x < \pi \\ 0 & \text{otherwise} \end{cases}$$

Soln:-

$$f(x) = \int_0^{\infty} (A(\omega) \cos \omega x + B(\omega) \sin \omega x) d\omega.$$

$$A(\omega) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(x) \cos \omega x \, dx$$

$$= \frac{1}{\pi} \int_0^{\pi} (\pi - x) \cos \omega x \, dx.$$

$$= \frac{1}{\pi} \left[(\pi - x) \frac{\sin \omega x}{\omega} - (-1) \left(-\frac{\cos \omega x}{\omega^2} \right) \right]_0^{\pi}$$

$$= \frac{1}{\pi} \left[-\frac{\cos \omega \pi}{\omega^2} - \left(0 - \frac{1}{\omega^2} \right) \right]$$

$$= \frac{1}{\pi} \left[\frac{1 - \cos \omega \pi}{\omega^2} \right]$$

$$B(\omega) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(x) \sin \omega x \, dx.$$

$$= \frac{1}{\pi} \int_0^{\pi} (\pi - x) \sin \omega x \, dx$$

$$= \frac{1}{\pi} \left[(\pi - x) \left(-\frac{\cos \omega x}{\omega} \right) - (-1) \left(-\frac{\sin \omega x}{\omega^2} \right) \right]_0^{\pi}$$

$$= \frac{1}{\pi} \left[\left(0 - \frac{\sin \omega \pi}{\omega^2} \right) - \left(\frac{-\pi}{\omega^2} - 0 \right) \right]$$

$$= \frac{1}{\pi} \left[\frac{\pi - \sin \omega \pi}{\omega^2} \right]$$

$$\begin{aligned}
 f(x) &= \int_0^{\infty} \left[\frac{(1 - \cos w\pi)}{\pi w^2} \cos wx + \frac{(\pi - \sin w\pi)}{\pi w} \sin wx \right] dw \\
 &= \frac{1}{\pi} \int_0^{\infty} \frac{(1 - \cos w\pi) \cos wx + (\pi - \sin w\pi) \sin wx}{w^2} dw
 \end{aligned}$$

20) (a) Find the Fourier sine transform of

$$f(x) = \begin{cases} x & 0 < x < 1 \\ 3-x & 1 < x < 3 \\ 0 & x > 3 \end{cases}$$

Soln :-

$$\begin{aligned}
 f_s^{\wedge}(\omega) &= \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \sin \omega x \, dx \\
 &= \sqrt{\frac{2}{\pi}} \left[\int_0^1 x \sin \omega x \, dx + \int_1^3 (3-x) \sin \omega x \, dx + \int_3^{\infty} 0 \, dx \right] \\
 &= \sqrt{\frac{2}{\pi}} \left\{ \left[x \left(-\frac{\cos \omega x}{\omega} \right) - (1) \left(-\frac{\sin \omega x}{\omega^2} \right) \right]_0^1 + \left[(3-x) \left(-\frac{\cos \omega x}{\omega} \right) - (-1) \left(-\frac{\sin \omega x}{\omega^2} \right) \right]_1^3 \right\}
 \end{aligned}$$

$$\begin{aligned}
 &= \sqrt{\frac{2}{\pi}} \left[\left(-\frac{\cos \omega}{\omega} + \frac{\sin \omega}{\omega} \right) + \right. \\
 &\quad \left. \left(0 - \frac{\sin 3\omega}{\omega} \right) - \left(-\frac{2 \cos \omega}{\omega} - \frac{\sin \omega}{\omega} \right) \right] \\
 &= \sqrt{\frac{2}{\pi}} \left[\frac{2 \sin \omega}{\omega} + \frac{\cos \omega}{\omega} - \frac{\sin 3\omega}{\omega} \right]
 \end{aligned}$$

(b) Represent $f(x) = e^{-kx}$, $x > 0$, $k > 0$ as a Fourier Cosine Integral.

Soln :-

$$f(x) = \int_0^{\infty} A(\omega) \cos \omega x \, d\omega \quad \text{where}$$

$$A(\omega) = \frac{2}{\pi} \int_0^{\infty} f(x) \cos \omega x \, dx$$

$$= \frac{2}{\pi} \int_0^{\infty} e^{-kx} \cos \omega x \, dx$$

$$= \frac{2}{\pi} \left[\frac{e^{-kx}}{k^2 + \omega^2} (-k \cos \omega x + \omega \sin \omega x) \right]_0^{\infty} = \frac{2k}{\pi(k^2 + \omega^2)}$$

$$\therefore f(x) = \int_0^{\infty} \frac{2k}{\pi(k^2 + \omega^2)} \cos \omega x \, d\omega$$